



3.2 Otitis Media

3.2.1 Acute Otitis Media

◆ **Key Features**

- Acute otitis media occurs when there is a bulging tympanic membrane with middle ear fluid.
- It is one of the most common childhood illnesses.
- It may be accompanied by hearing loss, nausea, vomiting, otorrhea, and fever.
- First-line treatment is amoxicillin. Watchful waiting is advocated in some patients.

Acute otitis media (AOM) is defined as middle ear inflammation lasting up to 3 weeks, accompanied by pain and bulging tympanic membrane (TM). Severe cases of AOM can be associated with high fever ($\leq 39^{\circ}\text{C}$). The TM can spontaneously rupture, and purulent otorrhea can occur. AOM should be differentiated from otitis media with effusion (also known as serous otitis media), mucoid otitis media (“glue” ear), and chronic otitis media. The mainstay of treatment is oral antibiotics, but some children with recurrent AOM or complications associated with AOM may require surgical management with tympanostomy tubes.

◆ Epidemiology

Sixty percent of all children under the age of one year will suffer at least one episode of AOM. That percentage increases to 85% by age 3. In children less than 10 years, AOM accounts for up to 20% of all clinic visits. Risk factors associated with AOM are daycare attendance, exposure to secondhand smoke, food or environmental allergies, and chronic rhinitis. Children between 3 and 18 months of age are the most susceptible.

◆ Clinical

Signs

The most important diagnostic feature for AOM is a bulging or full and opaque TM associated with middle ear effusion. A ruptured TM can present with purulent otorrhea. If acute mastoiditis is present, the postauricular region may also be erythematous and edematous; the auricle may also protrude anteriorly. Fever can also occur.

Symptoms

AOM is more common in children than in adults. It can present with otalgia, otorrhea, irritability, fever, nausea, vomiting, and feeding difficulties. Children may also tug on the ear. Hearing loss, ear fullness, and tinnitus can also be present. An upper respiratory tract infection may precede onset of AOM.

Differential Diagnosis

The differential diagnosis of AOM is a differential diagnosis for ear pain. Temporomandibular joint disease is a common cause of referred otalgia in adults. Other conditions include herpes zoster oticus, primary temporal bone malignancy, referred pain from other head and neck malignancies, nasopharyngeal mass causing auditory (eustachian) tube obstruction, cerumen impaction, cholesteatoma, otitis externa, mastoiditis, and foreign body. A common cause of otalgia in children is AOM; however, other pediatric causes include otitis externa, ear canal trauma, mastoiditis, and ear canal foreign body. It is important to differentiate AOM with otitis media with effusion (middle ear fluid without infection).

◆ Evaluation

Physical Exam

The most important portion of the physical exam is inspecting the TM. The auricle and the external canal remain normal in appearance and are not tender to palpation.

The appearance of the TM will change as the disease process follows its usual course. Initially, the TM is engorged and hyperemic. The hyperemia is most prominent along the manubrium of the malleus and the periphery of the drum. The TM is sluggish to pneumatic otoscopy, but all normal topographic landmarks are visible. As the infection progresses and the middle ear fills with purulence, the TM thickens, bulges, and loses normal landmarks. There may be erythema, tenderness, and edema in the postauricular region,

especially in small children. A tuning fork examination can demonstrate conductive hearing loss pattern.

If the infection progresses, the TM perforates in the pars tensa, and the patient experiences a resolution of pain and fever. If purulence is seen in the external canal, then cultures should be obtained. If the perforation heals and purulence reaccumulates, the infection may spread through the antrum into the mastoid; the mastoid trabeculae may begin to decalcify, leading to coalescent mastoiditis along with other complications. At this stage, the auricle displaces from the skull as postauricular edema increases. It is important to differentiate this from a severe otitis externa with painful cellulitis and swelling of the auricle.

Imaging

Imaging is usually not indicated unless coalescent mastoiditis or another complication of AOM is suspected. If a complication is suspected, a fine-cut temporal bone computed tomography (CT) scan is indicated.

Labs

Labs are rarely needed to treat routine AOM. Leukocytosis may be seen on complete blood count (CBC). Cultures can be obtained if the ear is draining, especially in infants less than 6 weeks of age. Routine tympanocentesis is not indicated in AOM but may be informative in patients who are immunocompromised or when suspicion for resistant or atypical pathogen is high.

Other Tests

An audiogram is not needed in the acute phase but can be helpful in evaluating children with recurrent AOM, especially in light of other cognitive delays. In cases where the presence of an effusion is in question, tympanograms will show low-volume flat (type B) or negative pressure (type C). Occasionally, the combination of a good pneumatic exam and an accurate tympanogram are needed to determine the presence of an effusion, even for an experienced otologist. Nasopharyngoscopy to evaluate for nasopharyngeal mass and auditory tube obstruction should be performed in adults with persistent unilateral middle ear effusion.

Pathology

Most commonly, AOM is caused by bacteria that gain entry into the middle ear cleft through the auditory tube. Infants are at higher risk for developing AOM because the auditory tubes are short and horizontal and feeding often occurs in the reclined position. As children grow, the auditory tubes become longer and more vertical, and the incidence of AOM declines. Other factors that contribute to AOM include allergic rhinitis, recurrent adenoiditis, adenoid (pharyngeal tonsil) hypertrophy, cleft palate, and immunodeficiency.

AOM in adults, especially if unilateral, may be ominous. The nasopharynx must be fully evaluated to identify the cause of the auditory tube blockage or dysfunction. A nasopharyngeal mass must be ruled out prior to attributing the cause to upper respiratory tract infection, allergic rhinitis, or recent flight.

Historically, the most common bacterial pathogens causing AOM have been *Streptococcus pneumoniae* (40–50%), nontypeable *Haemophilus influenzae* (20–30%), and *Moraxella catarrhalis* (10–15%). Vaccination with pneumococcal 13-valent conjugate vaccination (PCV13) can protect against many strains causing AOM. However, alterations in penicillin-binding proteins in *S. pneumoniae* and *H. influenzae* have increased the incidence of antibiotic resistance.

Histologically, middle ears display signs of inflammation and edema. The TM and middle ear mucosa are thickened and engorged with an inflammatory infiltrate. There is often frank purulence in the middle ear space.

◆ Treatment Options

Medical

There has been a trend toward withholding antibiotic treatment and advocating “watchful waiting” for 72 hours in children with uncomplicated AOM (< 24 months with unilateral AOM or ≥ 24 months with unilateral or bilateral disease). Pain is treated symptomatically. If AOM persists for more than 72 hours, the first-line antibiotic is amoxicillin (80 to 90 mg/kg/day) because it is effective, well tolerated, and inexpensive. A beta-lactamase antibiotic should be prescribed if there is a history of AOM within the last month, failed amoxicillin initial therapy, or conjunctivitis. If patients are penicillin-allergic, second- or third-generation cephalosporins, macrolides, or clindamycin are all options. Breastfeeding for initial 6 months of life, pneumococcal conjugate vaccine, and annual influenza vaccination can help prevent AOM. Reducing risk factors such as secondhand smoke exposure and daycare may also prevent AOM.

Otitis media with effusion (OME, also referred to as serous otitis media and nonsuppurative otitis media) is defined as the presence of middle ear fluid without signs or symptoms of acute ear infection. OME can occur after an episode of AOM; 75% of children with OME following AOM resolve by 3 months. When OME does not resolve spontaneously within 3 months, tympanostomy tubes should be offered.

Surgical

In patients in whom empiric therapy has failed, diagnostic tympanocentesis may be done for culture. Removing the fluid from the middle ear may also relieve pain. Tympanostomy tube placement may be offered if AOM is recurrent, which is defined as 3 episodes of AOM in the last 6 months or >4 episodes of AOM in the past 1 year. Tympanostomy tubes with adenoidectomy are recommended in children ≥ 4 years of age for OME lasting more than 3 months. For children < 4 years of age, adenoidectomy is performed only when there is nasal obstruction and chronic adenoiditis.

◆ Complications

Complications of tube placement include otorrhea, conductive hearing loss, retained tubes, cholesteatoma, and postextrusion TM perforations. Posttympanostomy otorrhea occurs in 10% of patients and is treated with

appropriate ototopical drops (i.e., fluoroquinolones, as they are not ototoxic). Most tubes will extrude spontaneously; however, removal of the tube may be necessary. TM perforations are estimated to occur 3 to 5% of the time and are followed conservatively. If they persist or cause considerable hearing loss, tympanoplasty may be recommended.

◆ Outcome and Follow-Up

Children with tympanostomy tubes require little postoperative care, and physician preference is usually the driving force determining postoperative recommendations. Many otolaryngologists advocate the use of 5 days of antibiotic topical drops twice daily, such as Ciprodex (Alcon Laboratories, Fort Worth, TX) or Floxin Otic (Daiichi Pharmaceutical Corporation, Montvale, NJ), postoperatively only to children with mucoid or purulent effusions found at the time of tube placement. Many otolaryngologists also recommend ear plugs for children only when swimming in fresh water and not during bath or shower time. The children are otherwise followed with a postoperative audiogram and interval visits at 6 to 12 months until the tubes extrude.

3.2.2 Chronic Otitis Media

◆ Key Features

- Chronic otitis media is long-term inflammation or infection affecting the middle ear space.
- Chronic otitis media may present with tympanic membrane perforation with otorrhea, retracted tympanic membrane with middle ear fluid, and cholesteatoma.
- The goal of medical and surgical treatment is to create a dry, safe ear.

Chronic otitis media (COM) comprises several long-term inflammatory or infectious disease processes that affect the middle ear. It can occur with or without a tympanic membrane perforation, cholesteatoma, and active drainage. When middle ear fluid exists, it may be serous (“chronic serous otitis media” or “chronic otitis media with effusion”) or thick (“chronic mucoid otitis media”). Purulent drainage through a tympanic membrane (TM) perforation is often referred to as “chronic suppurative otitis media.” COM can also be associated with retracted TM, granulation tissue, polyps, and ossicular erosion. Conductive hearing and otorrhea are common occurrences. Chronic auditory (eustachian) tube dysfunction can be the underlying cause in many patients. Treatment can include medical and surgical options.

◆ Epidemiology

The epidemiology of COM is not well defined. It affects males and females equally; however, there is a preference for certain populations: Native Americans, American Eskimos, and Australian Aborigines.

◆ Clinical

Signs

Signs of COM include perforated or retracted tympanic membrane, middle ear fluid, otorrhea, polypoid granulation tissue, ossicular erosion, and cholesteatoma. In patients with severe infection or cholesteatoma, facial palsy, nystagmus, or signs of meningitis can be seen in severe cases.

Symptoms

Symptomatology is broad and depends on the nature of the COM. Common symptoms include chronic or intermittent painless otorrhea and hearing loss. When COM is associated with cholesteatoma or severe infection, the otic capsule may be breached and sensorineural hearing loss, tinnitus, vertigo, dizziness, and imbalance can occur. The facial nerve can be involved, and facial droop may be reported. In severe cases, meningitis can occur and patients will present with altered mental status, fever, headache, and stiff neck. A chronic history of sinus disease and auditory tube dysfunction may also be elicited.

Differential Diagnosis

The differential diagnoses for chronic otorrhea and TM perforation are foreign body, temporal bone malignancy, Wegener's granulomatosis, and tuberculosis. Persistent polypoid tissue after medical treatment necessitates a biopsy. Wegener's granulomatosis may present as COM with fluctuating hearing loss and cranial nerve palsy without a previous history of OM. Tuberculous otitis media should be considered in a draining ear that does not improve despite maximal medical and surgical treatment. The classic description of tuberculous otitis media is painless otorrhea with multiple TM perforations.

◆ Evaluation

Physical Exam

A focused head and neck examination is performed, concentrating primarily on the otologic exam. The external auditory canal may be filled with moist debris that requires meticulous cleaning; at times, medical treatment of an inflamed ear canal is necessary to be able to visualize the TM. The TM should be inspected for retraction (e.g., myringoincudostapedioplasty), bulging, perforation, cholesteatoma, and granulation tissue. Debris and granulation tissue should be cleaned thoroughly with an otomicroscope. Abnormal tissue can be sent to pathology for review. The presence of middle ear fluid should be evaluated with pneumatic otoscopy. A tuning fork exam can indicate presence of conductive hearing loss. Cranial nerve examination,

particularly the facial nerve (CN VII), should be performed. A nasopharyngoscopy is recommended when there is unilateral COM with effusion in an adult patient in order to rule out nasopharyngeal mass.

Imaging

When COM is associated with cholesteatoma or intracranial or intratemporal complications, high-resolution computed tomography (CT) imaging of the temporal bones is recommended. Additional imaging such as magnetic resonance imaging (MRI) of the brain with and without gadolinium contrast is necessary when there are intracranial complications of COM. Uncomplicated COM may not require imaging.

Labs

Otorrhea, especially if recalcitrant to empiric therapy, should be cultured. Aerobic and fungal cultures are routinely sent. Additional blood work is rarely indicated. If the onset of symptoms is bilateral, recent, and accompanied by fluctuating hearing loss or cranial nerve palsy, then c-ANCA (cytoplasmic anti-neutrophil cytoplasmic autoantibody) is ordered to rule out Wegener's granulomatosis.

Other Tests

An audiogram is obtained once the ear is dry and prior to any surgical intervention. A conductive hearing loss is expected, and a mixed loss is not uncommon. Any granulation retrieved from the ear is sent to pathology to rule out malignancy.

Pathology

Historically, the most common pathogens were *Pseudomonas aeruginosa* and *Proteus*. Today, the most common pathogen is methicillin-resistant *Staphylococcus aureus* (MRSA). Pathogens gain access to the middle ear through the perforated TM. They then spread from the middle ear to the mastoid process. These same pathogens can also colonize the avascular debris collecting within a cholesteatoma. The middle ear mucosa becomes thick, fibrotic, and infiltrated with inflammatory cells. Mucosal edema leads to polyp formation and granulation. Bony vascular channels embolize secondary to chronic inflammation, leading to bone erosion, particularly involving the ossicular chain. Cholesteatoma may erode bone by additional local inflammatory response and osteolytic enzymes.

◆ Treatment Options

The goals of treatment are elimination of infection and restoration of function. The primary goal is to restore a "safe and dry ear."

Medical

COM may be initially treated with empiric ototopical antibiotic drops. Awareness of ototoxicity has made fluoroquinolone drops the preferred method of treatment. Ototopical drops reach the middle ear in such high

Table 3.4 Topical therapy for chronic otitis media

Topical agent	Typical dose regimen
2% acetic acid	5 drops 2–3 times daily
2% acetic acid/1% hydrocortisone solution	5 drops 2–3 times daily
Burow solution (13% aluminum acetate)	Drops or soaked pledget
Floxin (ofloxacin)* otic	10 drops twice daily
Ciprodex (ciprofloxacin/dexamethasone)† or Cipro HC (ciprofloxacin/hydrocortisone)‡ otic	5 drops twice daily
Cortisporin (neomycin/polymyxin B/ hydrocortisone)‡ otic suspension	5 drops 3 times daily
CSF powder (chloramphenicol 50 mg, p-aminobenzenesulfonamide 50 mg, and amphotericin 5 mg, with or without hydrocortisone 1 mg)	2 puffs twice daily
Gentian violet (aqueous)	Physician applies topically under microscope in office as needed
Boric acid powder	2 puffs twice daily

*Daiichi-Sankyo Co., Ltd., Tokyo, Japan.
†Alcon Pharmaceuticals, Inc., Fribourg, Switzerland.
‡Endo International, Dublin, Ireland.

concentrations that resistance is rarely an issue. See **Table 3.4** for topical treatment options. If the drainage does not respond, then cultures are indicated to rule out a resistant strain such as MRSA or a fungal infection. Vinegar washes or 2% acetic acid drops may be effective. There are several topical powders that also may periodically be applied if drops do not work. One such mixture includes ciprofloxacin, boric acid, dexamethasone, and fluconazole. Appropriate long-term IV antibiotics may also be indicated if an osteitis is suspected. Another effective topical powder preparation to help dry the chronically draining ear that is unresponsive to drops consists of chloramphenicol 50 mg, *p*-aminobenzenesulfonamide 50 mg, and amphotericin 5 mg, with or without hydrocortisone 1 mg; this is mixed and delivered in 1 or 2 puffs via a powder insufflator (e.g., Sheehy–House insufflator Otomed, Grace Medical, Memphis, TN) twice daily. Another option for office management is aqueous gentian violet, which has antifungal properties and may be “painted” over inflamed areas under the otomicroscope.

Surgical

The goals of surgery for COM are to eradicate the infection, improve hearing, prevent otorrhea, and remove cholesteatoma. Medical intervention with systemic or topical antibiotic preparations may be necessary to dry the ear prior to surgery. Surgical algorithms vary by preference, training, and experience. The following is a general list of options (not exhaustive) with definitions.

1. *Myringotomy* with tube placement is performed to remove middle ear fluid.

2. *Tympanoplasty without mastoidectomy* is performed to eradicate disease limited to the middle ear. Middle ear cholesteatoma can be removed through this technique, and the TM can be grafted if there is a TM perforation. TM grafting material includes loose areolar tissue, temporalis fascia, perichondrium, vein, or cartilage. The graft can be placed either medial or lateral to the TM.
3. *Atticotomy* can be performed in addition to tympanoplasty by removing the posterior bony annulus and scutum to better visualize the epitympanum and posterior mesotympanum. Through this approach, the entire ossicular chain can be visualized and cleaned, as well as the tympanic portion of the facial nerve. This can facilitate removal of granulation, retractions, and small cholesteatomas, especially for disease lateral to the ossicles. This is ideal for pars flaccida cholesteatoma that is lateral to the ossicular chain. Cartilage is used to reconstruct the scutum to prevent future retraction, and a formal mastoidectomy can be avoided.
4. *Tympanomastoidectomy with an intact canal wall* requires mastoidectomy and tympanoplasty to remove cholesteatoma or disease in the middle ear and antrum. A partial transcanal atticotomy may be performed if additional exposure is needed in the epitympanum. A facial recess approach can aid in removing disease in the sinus tympani and improve aeration. Microsurgical techniques are required to exenterate all involved air cell tracts. Removal of the malleus head is often required for disease extending into the anterior epitympanum or supratubal recess. A second look operation is often planned in cases of cholesteatoma, to assess for recurrent disease and/or perform ossicular reconstruction.
5. *Tympanomastoidectomy with canal wall down* is a technique that is employed for extensive cholesteatoma and rarely necessary in uncomplicated COM without cholesteatoma. This technique is also indicated in otherwise unresectable disease, noncompliant patients, only hearing or deaf ears, and patients with small or contracted/sclerotic mastoids prior to surgery. Removing the posterior canal wall increases visualization of the anterior epitympanum and the posterior mesotympanum and provides access to remove disease more effectively from the sinus tympani. Postmastoidectomy cavities may fail, harbor cholesteatoma, or chronically drain if a high facial ridge is left, there is a dependent mastoid tip, the TM graft fails, or the meatoplasty is inadequate. Surgeons must carefully attend to these fine details to ensure success.

◆ Complications

Complications of COM itself are divided into intratemporal and intracranial subcategories, and are covered in **Chapter 3.2.3**. Complications from surgery for COM are covered here. Postoperative complications are rare if surgery is performed meticulously. Postoperative pain is usually mild, and all patients calling with severe pain need to be evaluated. *Pseudomonas* is the most common offending organism in COM, and empiric treatment with fluoroquinolones is often effective. *Staphylococcus* is also a common pathogen for COM. When possible, wound infections are treated with

culture-directed antibiotics. Postoperative hematomas at the site of graft harvesting are rare but need to be drained when encountered.

Facial nerve injuries are rare. The use of facial nerve monitoring during even routine ear surgery is becoming widely accepted. Intraoperative injury needs to be addressed immediately with appropriate nerve exploration, decompression, and grafting if indicated. Getting the advice of a partner or a more experienced otologic surgeon can be helpful in these cases. Facial function that is normal initially and worsens over time is usually the result of edema; this is commonly treated with a two-week course of systemic steroids. When a patient awakens from a mastoid procedure with an unexpected facial weakness, it is prudent to observe the patient for a few hours to allow for the reversal of local anesthetic effects on the facial nerve prior to reexploration.

Cerebrospinal fluid (CSF) leak can be encountered intraoperatively and should be repaired intraoperatively. This can be done with a combination of fascia, cartilage, and/or bone cement. A neurosurgical consultation with lumbar drain placement is rarely required for small CSF leaks.

Mild vertigo is common following any middle ear procedure, but intraoperative labyrinthine injury results in severe postoperative vertigo and sensorineural hearing loss. Addressing the defect immediately with bone wax or fascia is recommended when the semicircular canal has been violated; patients may require hospitalization for symptomatic control of severe vertigo. Perilymphatic fistula may also present similarly. Oral or parenteral vestibular suppressants and antinausea medications are administered short term. Vestibular exercises can hasten recovery.

Granulations and synechiae in the ear canal or canal-wall-down mastoidectomy cavity are removed when encountered. Meatoplasties may require revision.

◆ Outcome and Follow-Up

Perioperative antibiotics are commonly administered prior to surgery for COM. The use of postoperative systemic antibiotics is controversial; however, active infection at the time of surgery necessitates systemic antibiotics, preferably culture-directed antibiotics. Surgery for COM is predominantly outpatient unless intracranial complications of COM are present. A variety of packing materials are utilized in the ear canal and canal-wall-down mastoidectomy cavities. Otological drops are prescribed to be administered twice a day. Although some packing can be removed 1 to 2 weeks postoperatively, the entirety of the packing is generally not removed for 4 to 6 weeks postoperatively to allow healing of graft. Routine postoperative audiometry is performed at 8 to 12 weeks. Success rates of surgical management for COM range from 78% to over 90% and likely are affected by disease severity and surgical technique. Cholesteatoma recurrence occurs 3 to 50% of the time. With these percentages in mind, patients are usually followed for several years.

3.2.3 Complications of Acute and Chronic Otitis Media

◆ Key Features

- Complications of acute and chronic otitis media may be intracranial, intratemporal, and extracranial.
- A decline in the incidence of otitis media complications is seen in the era of antibiotics.
- The most common extracranial complication is subperiosteal abscess; the most common intracranial complication is meningitis.
- Culture-directed antibiotics should be initiated when possible.

Complications of acute otitis media (AOM) and chronic otitis media (COM) are subdivided by site (extratemporal, intratemporal, and intracranial). When otitis media is associated with cholesteatoma, complications from cholesteatoma can also occur.

◆ Extratemporal Complications

Subperiosteal Abscess

Mastoiditis either directly erodes the bone of the lateral wall of the mastoid or traverses mastoid emissary veins into the subperiosteal space adjacent to the mastoid.

Signs and Symptoms

Fever, malaise, and otalgia are associated with a subperiosteal abscess.

Physical Exam

Otorrhea and an anteriorly and laterally displaced auricle can be seen on exam. The postauricular space can be erythematous, edematous, and fluctuant with fluid collection.

Imaging

Computed tomography (CT) of the temporal bones with contrast is recommended.

Treatment Options

Treatment is wide-field myringotomy or myringotomy with tube placement. The subperiosteal abscess should be incised and drained. A cortical mastoidectomy can be performed carefully if the mastoid is not severely inflamed. Culture should be obtained and empiric intravenous (IV) antibiotics started. Culture-directed antibiotics should be initiated when possible.

Bezold Abscess

Mastoiditis leads to bone erosion of the mastoid tip deep to the digastric ridge. Purulent material tracks into the neck, deep to the sternocleidomastoid muscle.

Signs and Symptoms

Fever, malaise, and neck pain are associated with a Bezold abscess.

Physical Exam

The patient may present with otalgia, otorrhea, postauricular tenderness, and a tender upper cervical mass.

Imaging

Contrast-enhanced CT will confirm the presence of a rim-enhancing cervical abscess in combination with mastoiditis.

Treatment Options

Recommended treatment includes IV antibiotics, wide-field myringotomy or myringotomy with tube placement, cortical mastoidectomy, and incision and drainage of the neck abscess. Culture-directed antibiotics should be initiated when possible.

◆ Intratemporal Complications

Labyrinthine Fistula

A labyrinthine fistula is caused by an erosion of otic capsule bone and exposure of the membranous labyrinth. This occurs in 7% of patients with cholesteatoma. Ninety percent of fistulas occur in the horizontal semicircular canal.

Signs and Symptoms

Most patients (62%) report periodic vertigo or imbalance. Patients may experience the Tullio phenomenon (dizziness induced by loud sounds) and sensorineural hearing loss. However, a significant number of patients will be asymptomatic.

Physical Exam

Fistula testing (nystagmus with pneumatoscopy) is positive in ~50% of patients. A tuning fork test may demonstrate sensorineural hearing loss.

Imaging

Thin-cut temporal bone CT is preferred; however, small fistulas may still be missed.

Treatment Options

Treatment of labyrinthine fistula is controversial because violation of the membranous labyrinth may result in a dead ear. Many authors recommend canal-wall-down mastoidectomy, leaving cholesteatoma matrix overlying

the fistula to form the lining of the exteriorized mastoid cavity. In cases of small fistulas that have not violated the membranous labyrinth, some authors advocate complete matrix removal and semicircular canal resurfacing with bone paste, fascia, or a similar sealant. Others recommend leaving the matrix in place, leaving the canal wall up, and coming back to remove the matrix and resurface the labyrinth at a second stage.

Coalescent Mastoiditis

Coalescent mastoiditis can occur in untreated or inadequately treated acute otitis media. Suppurative progression leads to erosion of the trabecular bone of the mastoid cavity. Time course of progression is typically 2 to 4 weeks.

Signs and Symptoms

Symptoms include fever, malaise, and otalgia.

Physical Exam

On the physical examination, purulent otorrhea or a bulging tympanic membrane (TM) can be seen. Postauricular erythema and edema, with displacement of the auricle anteriorly and laterally, is typical.

Imaging

A CT scan of the temporal bones demonstrates middle ear and mastoid opacification with erosion of trabecular mastoid bone. Erosion of the sigmoid sinus plate is not uncommon.

Treatment Options

Recommended treatment includes wide-field myringotomy or myringotomy with tube placement, IV antibiotics, and cortical mastoidectomy. Some advocate myringotomy with IV antibiotics for 3 to 6 weeks with follow-up CT scan to confirm resolution of infection. Culture-directed antibiotics should be initiated when possible.

Petrous Apicitis

Petrous apicitis is a rare complication resulting from the spread of infection into air cells within a pneumatized petrous apex (the prevalence of pneumatization is 30%).

Signs and Symptoms

The classic triad of deep retroorbital pain, purulent otorrhea, and ipsilateral abducens palsy (Gradenigo's syndrome) is seen. CN VII and VIII dysfunction also may occur.

Imaging

A CT scan of the temporal bones will show opacification of a pneumatized petrous apex. The trabeculae of the petrous apex may also be eroded. Contrast administration can demonstrate some rim enhancement consistent with abscess.

Other Tests

Magnetic resonance imaging (MRI) of the brain with and without gadolinium and lumbar puncture (LP) are helpful to assess concurrent intracranial processes.

Treatment Options

Empiric IV antibiotics should be initiated. Abscess, bone erosion, or failed medical therapy requires surgical drainage. Hearing status determines the choice of approach. Hearing preservation may be attempted by infracochlear, infralabyrinthine, retrolabyrinthine, subarcuate, and middle fossa approaches. Translabyrinthine or transcochlear approaches may be used for nonserviceable ears.

Facial Paralysis

Facial paralysis results from inflammation of dehiscence segments of the facial nerve secondary to infection.

Signs and Symptoms

Facial weakness may be sudden, progressive, complete, or partial.

Imaging

CT scan of the temporal bone to evaluate the integrity of the intratemporal facial (fallopian) canal and extent of cholesteatoma can be performed.

Pathology

In the setting of children with AOM, a congenitally dehiscence tympanic segment of the facial nerve is suspected. The paralysis is usually incomplete and resolves within 3 weeks. In the face of COM and cholesteatoma, dehiscence is typically due to facial canal erosion secondary to disease. Onset may be slow and progressive. The prognosis is poorer and recovery slower.

Treatment Options

Facial paralysis due to AOM is treated with wide-field myringotomy or myringotomy with tube placement, IV antibiotics, and systemic steroids. Cholesteatoma-associated paralysis requires mastoidectomy, nerve decompression proximal and distal to the diseased segment, and débridement of inflammatory tissue.

Acute Suppurative Labyrinthitis

Acute suppurative labyrinthitis results from direct bacterial invasion of the labyrinth, resulting in total auditory and vestibular loss. Acute suppurative labyrinthitis may lead to meningitis and vice versa.

Signs and Symptoms

Acute-onset total sensorineural deafness and severe vertigo are signs of acute suppurative labyrinthitis. Nausea and vomiting are common. Nystagmus can be seen on exam.

Pathology

Predisposing factors include congenital inner ear malformations and otic capsule erosion secondary to cholesteatoma.

Treatment Options

IV antibiotics are initiated and vertigo is treated symptomatically. Systemic steroid treatment is controversial. Residual dizziness and imbalance can be treated with vestibular therapy to promote central compensation. Wide-field myringotomy or myringotomy with tube placement is considered. Culture-directed antibiotics should be initiated when possible.

◆ Intracranial Complications

Meningitis

Meningitis is the most common intracranial complication of otitis media.

Signs and Symptoms

Fever, headache, nausea, vomiting, photophobia, and nuchal rigidity are symptoms of meningitis. Seizures, ataxia, and other focal neurologic signs are ominous findings.

Imaging

CT or MRI of the brain with and without contrast is essential to rule out other intracranial processes mimicking meningitis and to establish the safety of LP.

Other Testing

LP is diagnostic. Cerebrospinal fluid (CSF) can demonstrate increased pressure, increased protein, decreased glucose, inflammatory cells, or bacteria present. CSF culture is obtained.

Pathology

The potential routes of bacterial spread from the ear to the CSF space include hematogenous seeding, bony erosion with direct spread, and through bony channels (e.g., suppurative labyrinthitis, Hyrtl fissures, congenital inner ear defects, cochlear aqueduct, and traumatic defects).

Treatment Options

Because meningitis is the most common complication seen with otitis media, detailed antibiotic guidelines are listed in **Table 3.5**. First-line therapy is IV antibiotics and wide-field myringotomy or myringotomy with tube placement. Mastoidectomy is considered when medical therapy fails, there is radiographic confirmation of coalescent mastoiditis, or cholesteatoma is present. Use of systemic steroids is controversial but may reduce neurologic sequelae of meningitis and brain edema. Culture-directed antibiotics should be initiated when possible.

Table 3.5 Intravenous antibiotic treatment for meningitis*†

Neonate (<1 month) Ampicillin + gentamicin; or ampicillin + third-generation cephalosporin
Newborn (1–3 months) First choice: Ampicillin + (cefotaxime or ceftriaxone) + dexamethasone (0.15 mg/kg every 6 h × 4 d) Alternative: Chloramphenicol + gentamicin
Infant or child (>3 months) First choice: (Cefotaxime or ceftriaxone) + dexamethasone Alternative: Ampicillin
Older child or adult (Cefotaxime 2 g IV every 6 h or ceftriaxone 2 g IV every 12 h) + ampicillin 2 g IV every 4 h + dexamethasone 0.4 mg/kg every 12 h × 2 d + vancomycin (child 15 to 22.5 mg/kg IV every 6 h; adult 1 g IV every 12 h) If <i>Pseudomonas</i> suspected: Cefepime 2 g IV every 8 h instead of cefotaxime or ceftriaxone For a penicillin-allergic patient in whom cephalosporin cannot be used: Vancomycin 15–22.5 mg/kg IV every 12 h + trimethoprim/sulfamethoxazole 15 to 20 mg/kg/day IV divided every 6 h or Vancomycin 15 to 22.5 mg/kg IV every 12 h + chloramphenicol 1 g IV every 6 h ± rifampin 600 mg once daily

* Other regimens may be indicated for certain situations, such as an alcoholic patient, a patient with a CSF leak or a debilitating condition such as acquired immunodeficiency syndrome (AIDS), or a postneurosurgical patient.
† Length of treatment is generally at least 14 days.
(Data from Johns Hopkins Online Antibiotic Guide, <http://hopkins-abxguide.org>; and Greenberg MS. Handbook of Neurosurgery. 7th ed. Stuttgart/New York: Thieme, 2010.)

Brain Abscess

The most frequently lethal intracranial complication of OM is a brain abscess.

Signs and Symptoms

The clinical course is multistage, starting with fever, malaise, nausea, vomiting, headache, altered mental status, and seizures. This may be followed by a quiescent phase with moderate clinical improvement. The third stage, thought to represent abscess growth and ultimate rupture, is a rapid and fulminant return of symptoms with sudden clinical decline.

Imaging

CT brain or MRI brain with and without contrast is recommended. These studies can also aid in surgical planning.

Pathology

Infection spreads secondary to thrombophlebitis of venous channels leading from the mastoid to brain parenchyma. The temporal lobe and cerebellum are the most common sites of abscess.

Treatment Options

Treatment is IV antibiotics, neurosurgical consultation, and urgent craniotomy with drainage of abscess when medically stable. Wide-field myringotomy or myringotomy with tube placement should be performed. Mastoidectomy is typically recommended at the same time but may be delayed if the patient is medically unstable. Systemic steroids are controversial but may reduce brain edema and prevent neurological sequelae. Anticonvulsants are given if seizure risk is high. Frequent neurological checks and placement in the neurological intensive care unit are recommended. Culture-directed antibiotics should be initiated when possible.

Lateral Sinus Thrombosis

Bone overlying the sigmoid is eroded by infection, and perisinus inflammation leads to vessel wall necrosis and mural thrombus formation. Thrombus may propagate proximally to the confluence of sinuses and superior sagittal sinus, resulting in life-threatening hydrocephalus. Clot may also propagate distally into the internal jugular vein, leading to possible pulmonary embolus. Additionally, the infected thrombus may release septic emboli, leading to septicemia or deeper intracranial infections.

Signs and Symptoms

Patients report headache, malaise, and “picket-fence” pattern of diurnal spiking fevers. Neck pain may imply clot propagation distally into the neck. A Tobey-Ayer or Queckenstedt test (lack of CSF pressure elevation with occlusion of the internal jugular vein ipsilateral to the thrombosis) is suggestive of the diagnosis. Signs suggestive of hydrocephalus include papilledema on ophthalmologic exam and obtundation.

Imaging

Contrasted CT may show the “delta sign” resulting from sinus wall enhancement with a central filling defect. MRI and magnetic resonance venography (MRV) are very sensitive.

Treatment Options

Treatment is IV antibiotics, wide-field myringotomy or myringotomy with tube placement, and mastoidectomy. The sigmoid sinus and lateral sinus are surgically decompressed at the time of mastoidectomy. Inspection of the patency of the sigmoid sinus can be performed with a small-gauge needle and syringe. Incision into the thrombosed sigmoid sinus and removal of the infected thrombus can be considered. Ligation of the internal jugular vein should be performed if there is significant distal propagation and cervical infection. Anticoagulation is controversial but advocated when there is propagation of the thrombus towards the confluence of sinuses, cavernous sinus, or distally into the internal jugular vein. Culture-directed antibiotics should be initiated when possible.

Epidural Abscess

Granulation tissue and abscess form between the inner cortical bone of the skull base and adjacent dura.

Signs and Symptoms

Symptoms include headache, fever, and malaise. There are no specific symptoms or signs on exam; however, many patients report a deep mastoid pain.

Imaging

Contrast-enhanced MRI or CT of the brain will demonstrate a biconvex peripherally enhancing epidural collection adjacent to the temporal bone.

Treatment Options

IV antibiotics are initiated, and a wide-field myringotomy or myringotomy with tube placement in addition to mastoidectomy are performed. Depending on the location of the epidural abscess, decompression of the middle or posterior cranial fossa bony plate will drain the abscess. Granulation tissue can be débrided off of bone and intact dura. Neurosurgical consultation for burr hole placement or craniotomy to drain the abscess is recommended for collections that cannot be approached with a mastoidectomy. Culture-directed antibiotics should be initiated when possible.

Subdural Empyema

A subdural empyema is a rapidly progressive, fulminant bacterial infection between the dura and arachnoid layers. Infection spreads rapidly, resulting in significant brain edema, herniation, and death.

Signs and Symptoms

Presentation is similar to meningitis. Patient can present with fever, headache, altered mental status, seizures, and rapid deterioration. Progression of the empyema may result in more focal neurologic signs.

Imaging

Contrast-enhanced MRI or CT of the brain will demonstrate a crescent-shaped, peripherally enhancing collection adjacent to the bone.

Other Tests

LP is contraindicated because associated brain edema puts patients at high risk of tonsillar herniation.

Treatment Options

Treatment includes IV antibiotics, wide-field myringotomy or myringotomy with tube placement, neurosurgical consultation, and urgent neurosurgical drainage of the purulent collection. Mastoidectomy can be performed in conjunction with the craniotomy if the patient is hemodynamically stable. Culture-directed antibiotics should be initiated when possible.

Otitic Hydrocephalus

Otitic hydrocephalus is a condition of increased intracranial pressure associated with temporal bone infection.

Signs and Symptoms

The symptoms include headache, nausea, vomiting, blurry vision, diplopia, and lethargy. Papilledema or abducens palsy may be observed on eye exam.

Imaging

MRI or MRV will detect dural sinus thrombosis, rule out other intracranial causes, and confirm the safety of LP. In contrast to typical hydrocephalus, the ventricles are not dilated.

Pathology

The pathophysiology is controversial and is thought to be secondary to clot propagation into the confluence of sinuses and ultimate blockage of CSF flow through the arachnoid villi of the superior sagittal sinus.

Treatment Options

Treatment of lateral sinus thrombosis is imperative and was discussed. The otitic hydrocephalus is treated medically with acetazolamide, fluid restriction, and steroids. If medical management fails, lumbar drainage or long-term ventriculoperitoneal shunting may be considered. Culture-directed antibiotics should be initiated when possible.

3.2.4 Cholesteatoma

◆ Key Features

- A cholesteatoma is a slow-growing destructive lesion of the middle ear and mastoid.
- It consists of trapped and frequently infected desquamated keratin.
- A cholesteatoma requires definitive surgical management to avoid complications.

A cholesteatoma is a slow growing and destructive cystic lesion composed of desquamating keratin. It is accurately referred to as a keratoma. There are two types: acquired and congenital cholesteatoma. Acquired cholesteatoma is generally located superiorly or posteriorly in the middle ear and extends over time into the mastoid. A congenital cholesteatoma is generally found in the anterosuperior portion of the middle ear. Growth and infections of cholesteatomas can lead to intratemporal, extratemporal, and intracranial complications. Routine débridement of a dry and limited cholesteatoma pocket can be performed; however, definitive surgical resection is advocated for most patients who tolerate general anesthesia.

◆ Epidemiology

The prevalence or incidence of acquired cholesteatoma in the general population is unclear. The incidence of congenital cholesteatoma is 0.12 per 100,000 population. One to three percent of children with chronically draining ears have cholesteatoma.

◆ Clinical

Signs

Signs of a congenital cholesteatoma include conductive hearing loss and a white pearl in the anterosuperior quadrant of the middle ear behind an intact tympanic membrane (TM). Signs of an acquired cholesteatoma include conductive hearing loss and a posterior superior retraction pocket with a keratinaceous crust or desquamating debris. An aural polyp, granulation, or purulent otorrhea can also be seen. There may also be erosion of the ossicles, scutum, and posterior bony wall of the external auditory canal.

Symptoms

The symptoms of cholesteatoma are similar to that of chronic otitis media (COM). The most common presenting symptoms are painless chronic otorrhea and hearing loss. The otorrhea may be bloody or purulent. Patients may also present with complications of cholesteatoma, including dizziness secondary to a labyrinthine fistula, facial nerve paralysis, or fever, malaise, and altered mental status suggesting an intracranial process.

Differential Diagnosis

Cholesteatoma frequently presents with signs of COM, but not all patients with COM harbor cholesteatoma. The symptoms and the complications of both diseases are similar, and the two sometimes can be differentiated only in the operating room (OR). The differential diagnosis also includes benign and malignant neoplasms of the temporal bone and foreign body.

◆ Evaluation

Physical Exam

A general head and neck exam is performed, paying particular attention to the otologic exam and cranial nerves (particularly the facial nerve). An otomicroscope aids in meticulous cleaning and examination of the ear. Frequently, a large attic or posterior TM retraction with keratin debris is encountered. The retraction may be covered by a crust, which must be removed to complete the exam. Occasionally, a polyp may be encountered, covering the retraction and debris. The polyp should be manipulated with suction to see around it or debulk it. Polyps should *not* be aggressively removed with cup forceps, as ossicular injury can result. A congenital cholesteatoma will slowly grow undetected, but may be seen as a white “pearl” anterosuperiorly behind an intact TM. Nystagmus (Tullio phenomenon) can be seen with pneumatic otoscopy if a labyrinthine fistula is present. A tuning fork can indicate conductive or sensorineural hearing loss.

Imaging

Computed tomography (CT) is the initial imaging modality to evaluate the extent of cholesteatoma and the state of the bony structures of the temporal bone. A thin-section (1-mm) CT with true axial and coronal cuts is best. The cholesteatoma appears as a homogeneous soft tissue density that is frequently difficult to differentiate from soft tissue edema or fluid. Therefore, it is imperative to combine physical exam findings with CT results to diagnose cholesteatoma. In certain circumstances, magnetic resonance imaging (MRI) can be obtained with diffusion-weighted sequences to differentiate between cholesteatoma and other temporal bone pathology. Cholesteatomas restrict (hyperintense) on diffusion-weighted images.

Potential findings on coronal CT imaging:

1. Scutum erosion
2. Ossicular erosion
3. Tegmen dehiscence
4. Semicircular canal dehiscence or fistula
5. Dehiscence of the facial (fallopian) canal

Potential findings on axial CT imaging:

1. Ossicular erosion
2. Soft tissue opacification of antrum and air cells
3. Semicircular canal dehiscence or fistula
4. Dehiscence of the sigmoid sinus
5. Dehiscence of the facial canal

Other Tests

An audiogram with tympanogram is essential in determining the degree and nature of the hearing loss. Conductive hearing loss may suggest involvement of the TM and ossicular erosion/discontinuity.

Pathology

The pathophysiology of both congenital and acquired cholesteatoma remains controversial.

Pathology of Congenital Cholesteatoma

Congenital cholesteatoma is initially defined as a pearly mass behind an intact TM in the absence of otitis media (OM). The absence of otitis media was removed from the definition, as there is a high prevalence of OM in children. Two competing theories of formation are:

1. Epithelial rest: There is a localized epithelial rest that has been identified in fetal temporal bones at the lateral wall of the auditory (eustachian) tube in the anterosuperior quadrant of the middle ear that disappears at 33 weeks gestation. Persistence of this rest into childhood can account for congenital cholesteatoma.

2. Acquired inclusion: The speculation is that a microinjury and retraction occurs to the pediatric TM, allowing small foci of epithelial tissue to invade the middle ear and proliferate just anterior to the malleus manubrium.

Pathology of Acquired Cholesteatoma

There are four theories for acquired cholesteatoma.

1. TM retraction (primary acquired): There is retraction of the TM secondary to negative middle ear pressure from auditory tube dysfunction. This most commonly involves the pars flaccida retracting up into the attic, collecting keratinaceous debris, and forming a cholesteatoma.
2. Epithelial migration (secondary acquired): Migration of canal and TM epithelium through a TM perforation. This occurs most commonly with a marginal perforation. The epithelium then proliferates within the middle ear.
3. Implantation (secondary acquired): Similar to migration except that epithelium is actively implanted into the middle ear secondary to either trauma or surgery (e.g. myringotomy tube placement).
4. Metaplasia (historical theory): COM is believed to trigger metaplastic change of middle ear mucosa from cuboidal to keratinizing squamous cells.

In an actively growing cholesteatoma, there is local inflammation and infection. This inflammation causes fibroblasts and leukocytes to release enzymes that erode bone. Inflammatory cytokines also act as osteoclastic-activating factors that stimulate osteoclasts to erode bone. Persistent pressure by an expanding cholesteatoma may also erode adjacent bone.

◆ Treatment Options

Medical

There is no definitive medical therapy for cholesteatoma; however, the ear should be débrided with the aid of an otomicroscope, and coexisting infections should be treated with appropriate antibiotic/steroid ototopical preparations. Having a dry ear with low bacterial burden and granulation will aid in surgical treatment.

Surgical

The goals of surgery are to remove the disease, make a dry and safe ear, and restore hearing. Surgeons will differ on their methods of achieving these goals. Some surgeons prefer a staged method with ossicular chain reconstruction at a later stage; others prefer to do everything at a single stage (reexplore ears only with signs of persistent or recurrent disease). Surgical options follow. Postoperative antibiotics are usually recommended when infection is present at the time of surgery.

1. Tympanoplasty alone is performed to eradicate small cholesteatomas that are limited to the middle ear. The TM can be grafted if there is a TM perforation.

2. Atticotomy can be performed in addition to tympanoplasty by removing the posterior bony annulus and scutum to visualize the epitympanum and posterior mesotympanum better. Through this approach, the entire ossicular chain can be visualized and cleaned, as well as the tympanic portion of the facial nerve. This can facilitate removal of granulation, retractions, and small cholesteatomas, especially for disease *lateral* to the ossicles. This is ideal for pars flaccida cholesteatoma that is lateral to the ossicular chain. Cartilage is used to reconstruct the scutum to prevent future retraction, and a formal mastoidectomy can be avoided.
3. Tympanomastoidectomy with an intact canal wall requires mastoidectomy and tympanoplasty to remove cholesteatoma or disease in the middle ear and antrum. The bony ear canal is preserved. A partial transcanal atticotomy may be performed if additional exposure is needed in the epitympanum. A facial recess approach can aid in removing disease in the sinus tympani and improving aeration. A second-look operation can be performed to look for recurrent disease and perform a delayed ossicular reconstruction.
4. Tympanomastoidectomy with canal wall down (CWD) is a technique that is employed for extensive cholesteatoma, unresectable disease, noncompliant patients, disease in only hearing ears, largely eroded bony ear canals, cholesteatoma matrix adhering to a dehiscence facial nerve, contracted/sclerotic mastoids, and labyrinthine fistulas. Once the facial nerve is identified, the posterior wall of the ear canal is removed down to the level of the facial ridge. A low facial ridge will aid in postoperative mastoid débridement. All air cells are removed and all edges are saucerized. If there is a dependent mastoid tip, it can be amputated above the digastric ridge. The TM is grafted and ossicular chain reconstruction and meatoplasty are performed. Some of the mastoid cavity dead space can be obliterated with soft tissue from the postauricular region. Cholesteatoma recurrence rates are lower for CWD procedures, but patients require lifelong mastoid cavity care.
5. Radical mastoidectomy is a techniques that involves a CWD mastoidectomy, removing the TM and middle ear contents, and obliterating the auditory tube to render it nonfunctional.

◆ Complications

Complications from cholesteatoma and COM are addressed in a separate chapter (**Chapter 3.2.3**). Intraoperative complications in otologic surgery are prevented using careful technique with avoidance of any operative misadventures. Identifying solid landmarks such as the tegmen or lateral semicircular canal are important in preventing complications.

◆ Outcome and Follow-Up

Surgery is primarily outpatient; however, patients with suspected labyrinthine or cochlear fistulas remain overnight to monitor them for dizziness. Patients return for an office visit at either 1 or 2 weeks. At that

visit, most of the reachable packing is removed, and the patient is placed on ototopical antibiotic drops. CWD cavities are filled with dissolving packing or an antibiotic/steroid ointment, and ototopical drops are prescribed.

Patients are then seen at 4- to 6-week intervals to inspect the graft and clean any granulations. Water is kept out of the ears until complete epithelialization occurs. CWU ears can return to full preoperative activities once healed. CWD ears can engage in water-related activities with caution, knowing that they are at risk for infection secondary to retained moisture and dizziness from exposure to cold water. Cholesteatoma patients are followed for several years to monitor for recurrence. This involves yearly or twice-yearly visits for inspection and mastoid cavity cleaning if needed.

For surgeons who prefer a staged CWU technique, patients undergo a second-look operation within 6 to 12 months to rule out and manage recurrent or residual disease, and reconstruct the ossicular chain if indicated. If moderate disease is encountered, then a third look may be indicated; if extensive disease is encountered, a CWD procedure may be performed.

3.3 Otitis Externa

3.3.1 Uncomplicated Otitis Externa

◆ Key Features

- Otitis externa (OE) is an infectious process of the external auditory canal.
- The most common pathogens are *Pseudomonas aeruginosa* and *Staphylococcus aureus*.
- Otomycosis is less common and caused by *Aspergillus* and *Candida*.
- Treatment involves meticulous cleaning and topical preparations.
- Systemic therapy is required if the infection spreads out of the confines of the canal, or the patient is immunocompromised or a poorly controlled diabetic.

Otitis externa is a localized infection of the skin of the external auditory canal (EAC). The EAC contains varying amounts of cerumen and desquamated skin. Acute OE, called “swimmer’s ear,” is most common after water exposure but may also follow EAC trauma. Retained moisture will alkalize the canal, making it prone to bacterial infection. As long as the infection is confined to the ear canal, local aural toilet and topical drops will be curative. If the infection extends outside the confines of the canal to become a periauricular cellulitis or the patient has a complicating factor that may impede the effectiveness of topical antibiotics, then oral and occasionally IV antibiotics are required.

◆ Epidemiology

Acute otitis externa affects from 1:100 to 1:250 of the general population. A lifetime incidence may be as high as 10%. The disorder is more common in warm environments with high humidity and increased water exposure.

◆ Clinical

Signs

Otorrhea, conductive hearing loss, and EAC swelling are all common. As the skin of the EAC swells, the periosteum is irritated and becomes very painful. Pressing on the tragus or pulling the auricle may lead to significant pain. The auricle and periauricular tissues may also become edematous and tender if the condition becomes a periauricular cellulitis. If the infection has spread to a cellulitis, then the ear may be prominent with an increased auriculocephalic angle similar to that seen with acute mastoiditis. Signs of otomycosis include itching and visible hyphae on inspection.

Symptoms

Usually present with a 48- to 72-hour history of progressive pain, itching, discharge, and aural fullness. Patients may also complain of jaw pain. If the ear canal fills with debris or swells completely, then hearing loss will also occur.

Differential Diagnosis

The differential diagnosis includes foreign body of the EAC, otitis media, malignant or necrotizing OE in a diabetic, coalescent mastoiditis, malignancy, chronic OE, and other inflammatory lesions (e.g., eosinophilic granuloma). Dermatological conditions (eczema, contact dermatitis), allergic reaction to ototopical drops, and herpes zoster oticus with painful EAC vesicles need to be excluded. A localized furuncle may also mimic OE, as can chronic otitis media or acute otitis media with otorrhea. The presence of inflammation of the TM with formation of bullae and severe pain indicates bullous myringitis; this is rare, may be associated with influenza, and may be superinfected with bacteria. The best treatment for bullous myringitis is controversial but can include pain medications, oral antibiotics (e.g., macrolides), and ototopical antibiotic preparations.

◆ Evaluation

Physical Exam

A head and neck examination, concentrating on the otologic exam, is performed. When possible, the EAC should be débrided with an otomicroscope. Inspection may reveal eczema, otorrhea, edematous canal skin, erythema, moist cerumen, debris, or hyphae. One classic finding is pain with palpation or manipulation of the auricle or the tragus. Frequently, the skin is so swollen that the TM cannot be seen. If visible, the TM may appear inflamed but mobile (differentiating it from acute otitis media). The skin of the auricle and periauricular region may also demonstrate erythema, edema, and tenderness.

Imaging

Imaging is not needed for uncomplicated OE. However, when there is auricular or periauricular involvement with protruded auricle, a computed tomography (CT) scan of the temporal bones can help differentiate a coalescent mastoiditis from severe OE with postauricular cellulitis.

Labs

Labs are rarely indicated for uncomplicated OE. In patients with suspected uncontrolled diabetes, a blood glucose level may be helpful.

Other Tests

Cultures are recommended for recurrent, chronic, or recalcitrant OE. Cultures should be sent for routine culture and sensitivity as well as fungus. Any abnormal-appearing tissue or polyp that does not respond to medical management should be biopsied to exclude malignancy.

Pathology

The infection usually begins with moisture buildup in the EAC. The acidic and hydrophobic qualities of cerumen make it bacteriostatic. A warm, moist EAC with decreased cerumen favors bacterial overgrowth. The bacteria will readily invade the skin. Although > 90% of OE cases are bacterial, a moist alkaline environment also favors fungal growth. *Pseudomonas aeruginosa* and *Staphylococcus aureus* are most common bacteria, and *Aspergillus* and *Candida* are the most common fungi.

◆ Treatment Options

Medical

The ear must be meticulously cleaned with an otomicroscope with complete removal of debris. This may need to be repeated in a few days. Once the ear is cleaned, otic drops should be placed. There are several preparations available, including nonantibiotic, antibiotic alone, and antibiotic plus steroid therapies (**Table 3.6**). They may be used two or three times per day for 7 to 14 days. Dry ear precautions are recommended until the infection clears.

Placement of a wick (e.g., Pope Oto-Wick; Medtronic, Inc., Minneapolis, MN) to carry drops medially may be required, especially in canals so swollen that the TM is not easily visible. Keep in mind that placing a wick can be a very uncomfortable experience for the patient. Analgesics, even those with narcotic, may be required to control the pain. As edema regresses, the wick will usually fall out or may be removed. OE due to a foreign body will not resolve without removal of the foreign body.

Relevant Pharmacology

Topical preparations are superior to systemic medications in cost and side effect profile. The medication concentration is significantly higher with topical medications than with systemic drugs. The efficacy of ototopical drugs depends on drug delivery; therefore, ear débridement is critical and wicks are placed when appropriate.

Table 3.6 Topical preparations useful for management of otitis externa

Topical agent	Typical dose regimen
Antibacterial agents	
2% acetic acid	5 drops 3–4 times daily
2% acetic acid/1% hydrocortisone solution	5 drops 3–4 times daily
Burow solution (13% aluminum acetate)	Drops of soaked pledget
Floxin Otic (ofloxacin otic)*	5 drops twice daily
Ciprodex (ciprofloxacin/dexamethasone)† or Cipro HC (ciprofloxacin/hydrocortisone)† otic	5 drops twice daily
Cortisporin otic suspension (neomycin/polymyxin B/hydrocortisone)‡	3–5 drops 3 times daily
Antifungal agents	
Clotrimazole (1%)	4 drops 3–4 times daily
Gentian violet (aqueous)	Physician applies topically under microscope in office as needed
Boric acid solution (3%)	5 drops twice daily
Boric acid powder	1–2 puffs 1–2 times daily
CSF powder (chloramphenicol, p-aminobenzenesulfonamide, amphotericin ± hydrocortisone)	1–2 puffs 1–2 times daily

*Daiichi Pharmaceuticals, Montvale, NJ.

†Alcon Pharmaceuticals, Inc., Fribourg, Switzerland.

‡Endo International, Dublin, Ireland.

Note: Use in combination with serial débridement; may require Oto-Wick placement if canal is severely swollen.

Otological antibiotic drops include Cortisporin (neomycin/polymyxin B/hydrocortisone), ofloxacin, and ciprofloxacin. There is a 10 to 15% sensitization rate with neomycin containing products. Acidifying solutions such as Domeboro (2% acetic acid; Bayer Consumer Health, Morristown, NJ) or VoSoL (2% acetic acid and propylene glycol) can be used. Currently, U.S. Food and Drug Administration (FDA)-approved preparations for patients with tympanic membrane perforations and exposed middle ear are ofloxacin and ciprofloxacin/dexamethasone.

Surgical

There is no specific role for surgery in acute OE. Patients may occasionally require a general anesthetic to carry out the canal cleaning and wick placement if the ear is too tender to instrument in the office. If a foreign body is suspected, especially in a child, then a general anesthetic may be needed to remove it.

Chronic OE results after fibrosis thickens the skin of the EAC, sometimes to the point of completely obliterating the canal. In some of these cases, a canalplasty with meatoplasty and skin grafting may be necessary.

◆ Outcome and Follow-Up

Most cases resolve within 7 to 10 days of treatment. Causative factors such as eczema, swimming, or cotton swab (Q-tip) use need to be addressed to promote resolution and prevent recurrence. Patients may need to be seen frequently in the office for repeated cleanings until the infection resolves. Patients should keep the ears dry for 7 to 10 days. Swimmers can return to the water with waterproof ear plugs in 3 days. Hearing aids can be replaced after pain and discharge resolve.

3.3.2 Malignant Otitis Externa

◆ Key Features

- Malignant otitis externa (MOE) is most frequently caused by *Pseudomonas aeruginosa*.
- Patients are immunocompromised, most commonly with diabetes.
- It is characterized by severe deep aural pain, especially at night.
- The facial nerve is the most common cranial nerve involved.

Malignant otitis externa (MOE) is a severe and serious form of infective otitis externa (OE). It is also referred to as necrotizing otitis externa. Diagnostic criteria include pain, edema, exudate, granulation at the bony-cartilaginous junction, failed local medical treatment > 1 week, abnormal culture, and abnormal bone imaging studies. When diagnosis and treatment are delayed, MOE can progress to skull base osteomyelitis, cranial nerve palsies, and death. Medical treatment with long-term systemic antipseudomonal antibiotics is the mainstay of treatment. Surgical management has little role in treating MOE.

◆ Epidemiology

Ninety percent of patients are diabetic, and most are over 60 years old. Other predisposing factors include immunosuppression (e.g., human immunodeficiency virus [HIV], transplant patients, chemotherapy).

◆ Clinical

Signs

Patients will have external auditory canal (EAC) and periauricular tissue edema and tenderness. The tympanic membrane (TM) may be normal or inflamed with a middle ear effusion. There is granulation tissue along the floor of the canal at the bony-cartilaginous junction. Palsy of cranial nerves (CN) VII, IX, X, XI and XII may be present. Visual disturbance from CN III, IV, and VI palsies and facial paresthesias can also occur during the progression of skull base osteomyelitis.

Symptoms

Aural pain and discharge are not responsive to the usual measures. Pain is often severe at night and may awaken patients from sleep. Also, headache and temporomandibular joint (TMJ) pain are common. Symptoms typically persist for more than a week. As the disease progresses, patients may complain of facial weakness, dysphagia, hoarseness, shoulder/tongue weakness, lethargy, nausea, blurred vision, and mental confusion.

Differential Diagnosis

MOE must be differentiated from routine OE or acute otitis media with mastoiditis. Fungal MOE must also be considered. Granulation tissue should be biopsied to rule out a malignancy. The differential diagnosis also includes EAC foreign body, infected cholesteatoma, other inflammatory disorders (e.g., Wegener's granulomatosis), and other infectious disorders (e.g., tuberculosis).

◆ Evaluation

History

The history should focus on elucidation of predisposing factors, such as diabetes and immunosuppressive disorders (e.g., HIV, leukemia, chemotherapy, transplant, malnutrition, alcoholism). If the patient is diabetic, information regarding recent glucose levels and hemoglobin A1c (HbA1c) is useful.

Physical Exam

A complete head and neck examination concentrating on the cranial nerve and otologic exam is warranted. The ear canal is débrided and examined carefully using an otomicroscope. This may require general anesthetic if pain is severe. Parotid palpation and TMJ mobility are noted.

Imaging

High-resolution computed tomography (CT) of the temporal bone is recommended. Findings on CT can include bony EAC or skull base erosion, soft tissue thickening of the EAC, and opacification of the middle ear and mastoid. CT is not as useful for determining intracranial and soft tissue involvement nor for determining the endpoint of treatment (bony erosion persists on CT despite resolution of infection). Magnetic resonance imaging (MRI) of the skull base with and without gadolinium contrast is helpful in detecting the progression of disease and is superior to CT in detecting intracranial, dural, and soft tissue involvement.

Technetium-99m (^{99m}Tc) nuclear scanning highlights areas of osteoblastic activity seen in osteomyelitis. Increased uptake, indicative of skull base osteomyelitis, is seen a few hours after ^{99m}Tc administration. This modality may be positive prior to bony changes seen on CT but cannot be used to determine the end point of treatment (can remain positive for months to years as a sign of bony repair and remodeling). It is very helpful in confirming an early diagnosis.

Table 3.7 A consideration of key diagnostic tests for malignant otitis externa

Labs
• CBC with differential (follow ANC level if neutropenic) • Blood glucose level (serially) • HbA1c, ESR (serially)
Pathology
Histopathologic exam of granulation tissue to exclude neoplasm
Microbiology
Bacterial and fungal culture and sensitivities of ear canal debris
Imaging
• Temporal bone CT to assess for extent of disease, bone erosion • MRI with gadolinium may be helpful to assess dural, intracranial, and soft tissue involvement. • Nuclear scanning: ^{99m}Tc scan is better for early diagnosis of MOE and determining extent of skull base osteomyelitis; ⁶⁷Ga and ¹¹¹In nuclear scanning are preferred for follow-up and assessment of disease resolution.

Abbreviations: HbA1c, hemoglobin A1c; ANC, absolute neutrophil count; CBC, complete blood count; CT, computed tomography; ESR, erythrocyte sedimentation rate; MRI, magnetic resonance imaging.

Gallium-67 (⁶⁷Ga) or indium-111 (¹¹¹In) nuclear scanning highlights areas of inflammation by binding to leukocytes. Increased uptake, indicative of skull-base osteomyelitis, can be seen on imaging 24 to 72 hours after administration of the radioactive tracer. It is a good test for end point treatment because it reverts to normal once the infectious response has resolved. It may be repeated several times throughout the treatment course to monitor response.

Labs

Routine laboratory testing will display an increased white blood cell (WBC) count, an increased erythrocyte sedimentation rate (ESR), and elevated blood glucose (in diabetics). Serial ESR levels can be used to monitor treatment response.

Other Tests

The ear canal should be swabbed for culture and sensitivity. Granulation tissue should be biopsied to rule out malignancy. A baseline audiogram is important, especially when utilizing potentially ototoxic antibiotic medications and determining the effects of skull base osteomyelitis on the otic capsule. See **Table 3.7** for a consideration of diagnostic tests.

Pathology

Pseudomonas aeruginosa is the causative bacteria in a majority of cases; however, the incidence of methicillin-resistant *Staphylococcus aureus* (MRSA) has increased. Infection begins at the skin of the EAC and spreads into the mastoid and temporal bone directly or through venous channels and fascial planes. Infection can also spread to the parotid gland, infratemporal fossa, and skull base via the fissures of Santorini and the tympanomastoid suture line. Infection and inflammation involving adjacent bone can cause cranial nerve palsies.

Patients are immunocompromised; the most common reason for immunosuppression is uncontrolled diabetes mellitus. Their cerumen is neutral pH, which promotes bacterial overgrowth and infection. MOE from fungal etiology can be seen in HIV with acquired immune deficiency syndrome (AIDS), hematologic malignancies, and immunosuppression after transplantation. *Aspergillus* is usually the culprit.

◆ Treatment Options

Medical

Medical therapy is the mainstay of treatment; however, there are no established guidelines for treatment. Early, uncomplicated MOE from nonresistant *Pseudomonas aeruginosa* may be treated with oral fluoroquinolone in an outpatient setting with close follow-up. However, severe and complicated MOE require parenteral therapy with good bone penetration. Prior to the development of a third-generation antipseudomonal cephalosporin, empiric treatment with intravenous antipseudomonal penicillin and aminoglycoside was commonly recommended. Systemic therapies for MOE are described in **Table 3.8**. Culture-directed antibiotics are used when possible. Some advocate two systemic medications over monotherapy in order to prevent drug resistance.

Otological treatments are also initiated early in the disease process (e.g., fluoroquinolones ± steroid, neomycin/polymyxin B/hydrocortisone, tobramycin, gentamicin). When possible, nontotoxic drops should be prescribed when a TM perforation exists. Wick placement may be necessary when the EAC is narrow or completely stenosed.

Long-term oral or systemic therapy is recommended and can range from 4 weeks to 6 months. The end point of therapy for MOE is controversial, varies from institution to institution, and may rely on availability of resources. In general, treatment has been stopped when serial $^{67}\text{Ga}/^{111}\text{In}$ scan obtained every 4 to 6 weeks returns to normal. Alternatively, normalization of serial ESRs and three negative ear cultures has also been argued. Patients with skull base osteomyelitis may need to be on lifelong therapy, especially to prevent relapse. Failure of therapy may indicate development of multi-drug-resistant pathogens or surge of other pathogens; new cultures should be performed.

In addition, immunosuppression is reversed when possible (e.g., correcting diabetes, halting chemotherapy, initiating antiretroviral therapy for HIV). Hyperbaric oxygen is considered in patients with extensive skull

Table 3.8 Empiric antibiotic therapy for malignant otitis externa

Antibiotic	Class	Dosage
Ciprofloxacin	Fluoroquinolone	750 mg orally every 12 hours 400 mg IV every 12 hours
Ceftazidime	Third-generation cephalosporin	2 g IV every 8 hours
Cefepime	Fourth-generation cephalosporin	2 g IV every 12 hours
Piperacillin-tazobactam	Antipseudomonal penicillin	3.375 or 4.5 g IV every 6 hours
Ticarcillin-clavulanate	Antipseudomonal penicillin	3 g IV every 4 hours
Gentamicin	Aminoglycoside	1–1.66 mg/kg IV every 8 hours
Tobramycin	Aminoglycoside	1–2 mg/kg IV every 8 hours

Note: Aminoglycosides are usually prescribed along with an antipseudomonal antibiotic. Aminoglycosides require peak and trough monitoring and observation for renal failure and hearing loss. Therapy should be culture-directed whenever possible.

base osteomyelitis or progressive disease despite aggressive medical management.

Surgical

Traditionally, surgical management is limited to initial and serial canal débridement. A myringotomy tube is considered if there is middle ear fluid.

◆ Complications

Progressive MOE becomes osteomyelitis of the skull base. With spreading infection, CNs VII, IX, X, XI, and XII may become involved. Consequences include aspiration with difficulties in speech and swallowing. CNs III, V, and VI may also be involved. Meningitis, lateral sinus thrombosis, cerebral abscess, and death may all occur.

◆ Outcome and Follow-Up

With proper diagnosis and treatment, cure rates are over 80%. Mortality rates have significantly declined in the antibiotic era. Serial examinations and débridements in follow-up may be necessary. Topical and systemic treatment is generally completed when the TM and EAC return to normal appearance, pain resolves, cultures are negative, ESR normalizes, and there is a negative ⁶⁷Ga/¹¹¹In scan. Treating the underlying immunosuppression is an important factor in the management of MOE and skull base osteomyelitis.